

Laboratory assignment №2

Topic : Development with ASP.NET MVC. Project configuration. Displaying data from DB on web-pages.

Goal: Get acquainted with the MVC architecture, acquire skills in creating the main components of the MVC architecture, learn how to configure database connections and execute data reading queries, and acquire skills in working with views.

Workflow:

Task 1. Project creation and configuration

Create a project based on the MVC architecture following the instructions below

Number on group list	Topic	Short descriptions of functionalities
3	Job Portal	Employers can post job listings, and candidates can apply for jobs and manage their applications.

Fig. 1 – My task

I wrote

dotnet new globaljson --sdk-version 8.0 --output JobPortalSln/JobPortal
and created a global json file for possible future connections

Then I wrote in the console:

dotnet new web --no-https --output JobPortalSln/JobPortal --framework net8.0

and created the project

And made solution:

dotnet new sln -o JobPortalSln

The next one is:

dotnet sln JobPortalSln add JobPortalSln/JobPortal

This command adds the JobPortal project to the JobPortalSln solution file.

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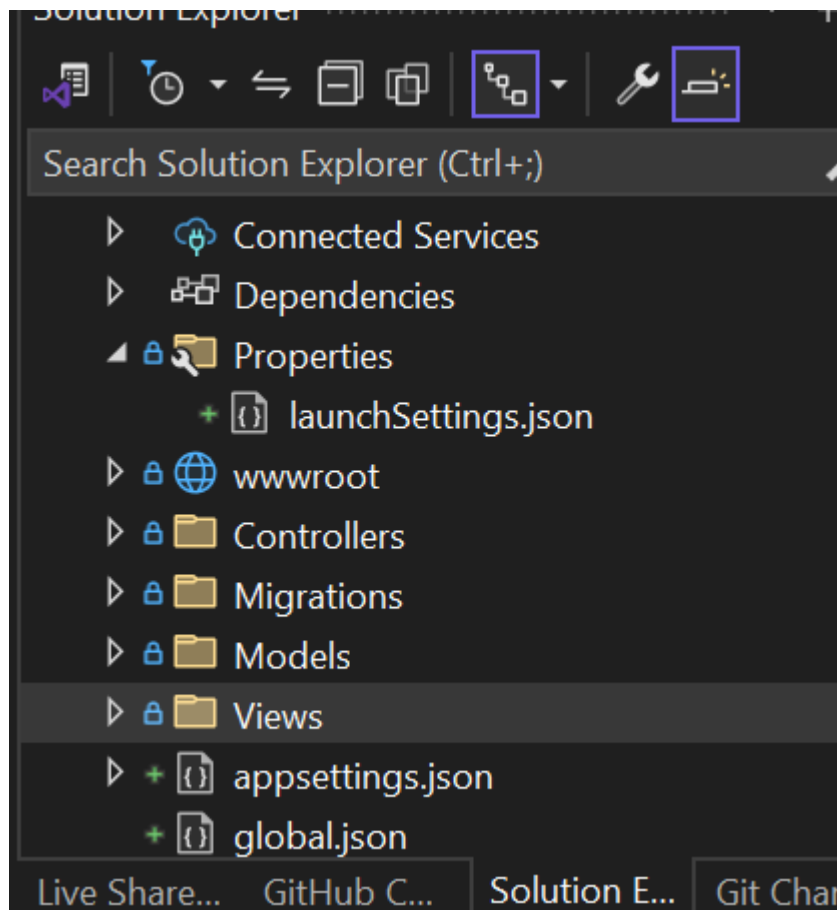


Fig. 2 – My solution explorer

Then I changed my launchsettings.json

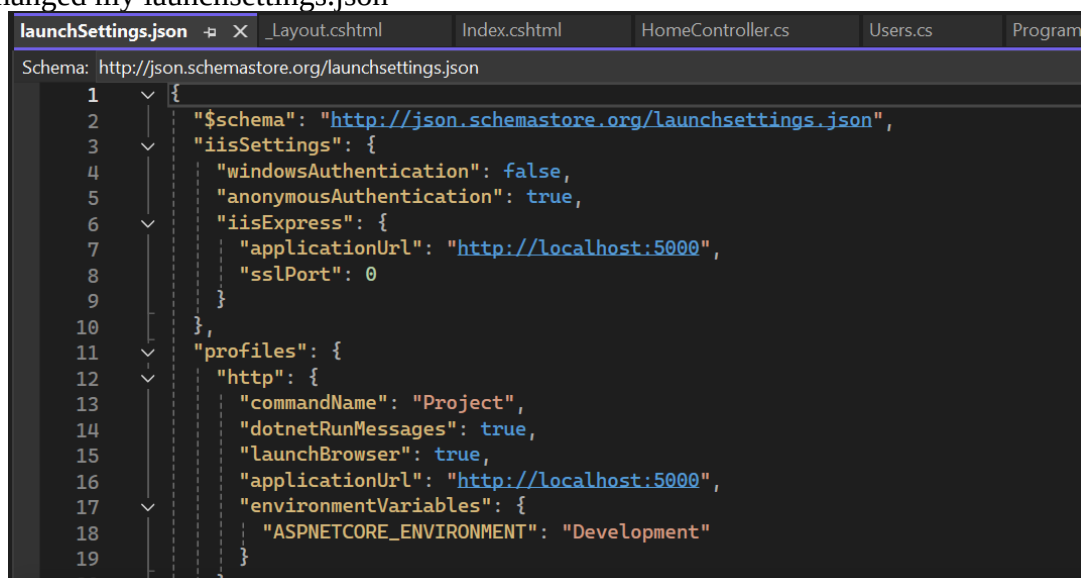


Fig. 3 – Wrote “channels” for host

Next I created the project structure with MVC principle (Models, Views, Controllers)

And also I wrote it in Program.cs:

```
var builder = WebApplication.CreateBuilder(args);
```

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```
builder.Services.AddControllersWithViews();

var app = builder.Build();

app.UseStaticFiles();

app.MapDefaultControllerRoute();

app.Run();
```

It's important, because we connected all configurations.

Added a Razor View file called `_ViewStart.cshtml` to the Views folder with the following content:

```
@{
    Layout = "_Layout";
}
```

The Razor View Start file tells Razor to use the layout file in the HTML it generates, reducing the number of duplicate views.

I added a class file called `HomeController.cs` to the `/Controllers` folder and implemented the class in it:

```
public class HomeController : Controller
{
    public IActionResult Index() => View();
}
```

To create the view, I added a Razor View file called `Index.cshtml` to the Views/Home folder with the content shown in the listing:

```
<h4>Welcome to JobPortal</h4>
```

Create a file with the appropriate name in the Models folder

```
using System.ComponentModel.DataAnnotations;
using System.ComponentModel.DataAnnotations.Schema;
```

```
namespace JobPortal.Models
{
    public class Users
    {
        public long? UserID { get; set; }
        public string Name { get; set; }

        public int Password { get; set; }
        public string Login { get; set; }
        public string PhoneNumber { get; set; }
        public string Description { get; set; } = String.Empty;
        public string City { get; set; } = string.Empty;
    }
}
```

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```

        public string Type_access { get; set; }
        public int? CompanyID { get; set; }

    }
}

```

After completing all the previous steps, I converted an empty WebAPI project into a project that implements the MVC pattern and returns ready-made HTML pages in response to requests.

Task 2. ORM setup

Configure the database using the instructions below

I already had the SQL Server Express 2019 LocalDB packages installed

Added the Entity Framework Core package to the project:

```

dotnet add package Microsoft.EntityFrameworkCore.Design
dotnet add package Microsoft.EntityFrameworkCore.SqlServer

```

Also added the necessary packages for working with migrations

```

dotnet tool install --global dotnet-ef

```

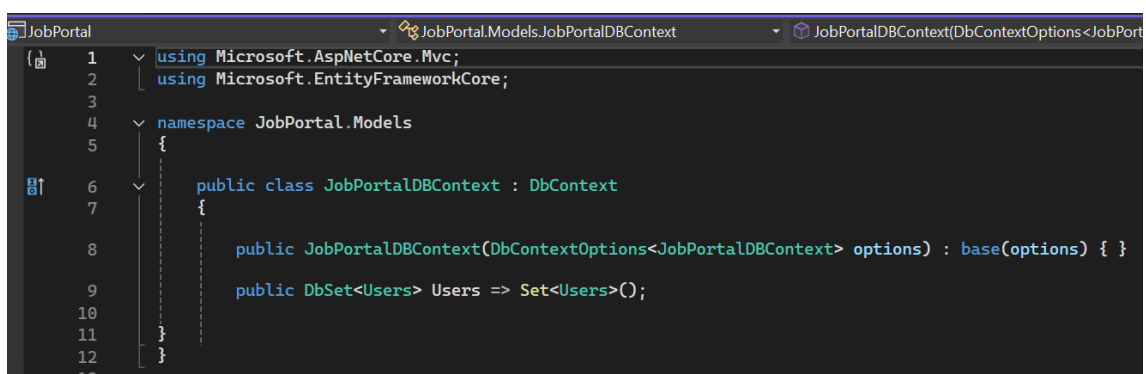
I added the connection string to my appsettings.json

```

{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "JobPortalConnection":
    "Server=(localdb)\\MSSQLLocalDB;Database=JobPortalDB;MultipleActiveResultSets=true"
  }
}

```

I also added the database context



```

1  using Microsoft.AspNetCore.Mvc;
2  using Microsoft.EntityFrameworkCore;
3
4  namespace JobPortal.Models
5  {
6      public class JobPortalDBContext : DbContext
7      {
8          public JobPortalDBContext(DbContextOptions<JobPortalDBContext> options) : base(options) { }
9          public DbSet<Users> Users => Set<Users>();
10     }
11 }
12
13

```

Fig. 4 – DB Context file

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After I did the migrations, a database was created (respectively on the localdb server which it was written in the appsettings.json)

```
PS D:\ASP_development\pr2_example\JobPortalSln\JobPortal> dotnet ef migrations add Initial
Build started...
Build succeeded.
Done. To undo this action, use 'ef migrations remove'
PS D:\ASP_development\pr2_example\JobPortalSln\JobPortal> dotnet tool install --global Microsoft.Web.LibraryManager.Cli
You can invoke the tool using the following command: libman
Tool 'microsoft.web.librarymanager.cli' (version '3.0.67') was successfully installed.
```

Fig. 5 – Doing migrations

I created SeedData.cs and filled the DB with three notes
Final result

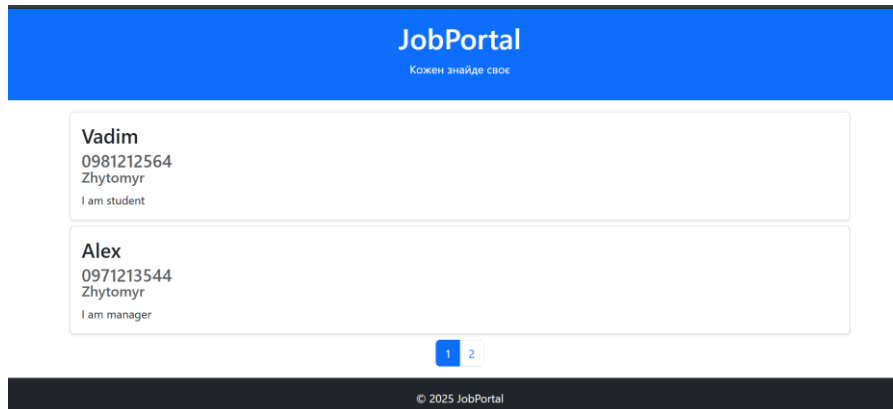


Fig. 6 – My main screen

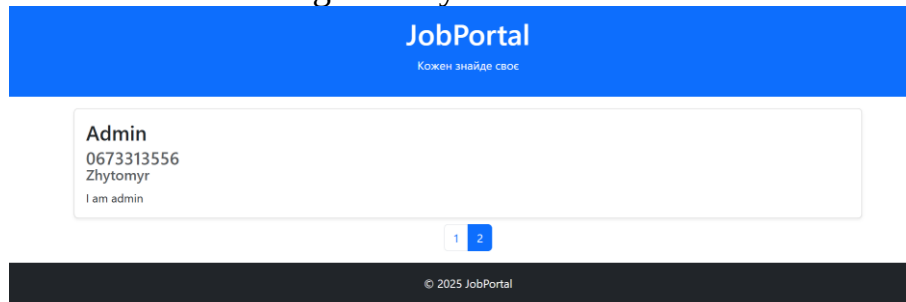


Fig. 7 – Second page

Conclusions: During the laboratory work, I Get acquainted with the MVC architecture, acquire skills in creating the main components of the MVC architecture, learn how to configure database connections and execute data reading queries, and acquire skills in working with views.

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